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# Feedback-Based Shape Control of Deformable Linear Objects via a Simplified Model in Robotic Manipulation

Research Practice – Preliminary Note

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Rémy Tribout · Technical University of Munich · May 11, 2026

## Motivation

This work is motivated by the difficulty of accurately modeling and controlling deformable linear objects (DLOs), such as cables or rods, which are frequently encountered in robotic manipulation tasks. DLOs cannot be easily handled using classical rigid-body assumptions, due to their nonlinear continuum mechanics and high-dimensional configuration space.

Recent studies have addressed this problem through different modeling and control strategies. Physically grounded approaches based on Cosserat rod theory provide accurate continuous descriptions of deformation and have been successfully used in optimization-based and closed-loop control frameworks for robotic manipulation [1, 2]. In contrast, simplified models such as mass-spring-damper systems have been widely used in robotic simulation due to their computational efficiency, despite reduced physical fidelity [3]. From an application perspective, robotic co-manipulation strategies have demonstrated the relevance of DLO manipulation in practical scenarios, highlighting the need for reliable and efficient modeling approaches [4].

These works highlight a trade-off between model accuracy and computational efficiency in DLO manipulation. High-fidelity models enable accurate control but are often computationally expensive, while simplified models are efficient but less accurate. This gap motivates feedback-based correction strategies to improve the predictive performance of simplified models using higher-fidelity references.

This research practice focuses on this model mismatch problem and aims to develop a feedback-based correction framework for improving simplified deformable object models in the context of robotic manipulation.

## Preliminary Work

A first implementation of a deformable object simulation framework has been developed based on the Cosserat rod model presented in [1]. The model was implemented in Python and numerically integrated to simulate the static deformation of a slender rod under applied boundary forces and moments.

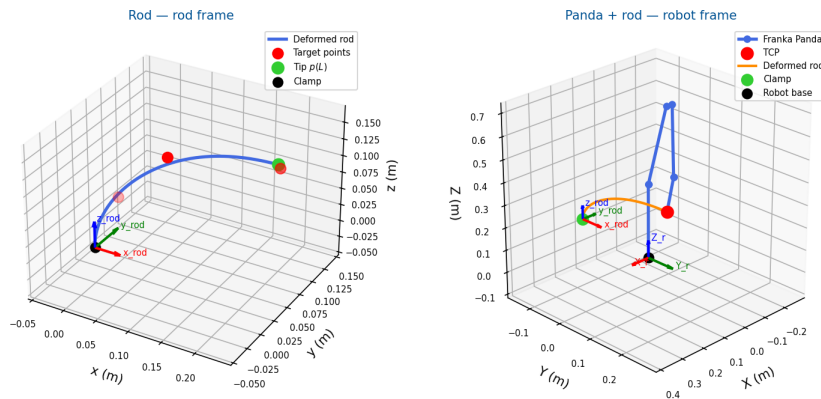


Figure 1: Cosserat-based rod deformation with optimized shape matching and corresponding robot configuration

An optimization-based method was used to compute the actuation required to match a desired target shape (set of points). A continuation strategy was implemented to improve convergence from the initial straight configuration to the target deformation.

In addition, a robotic manipulator model (Franka Emika Panda) was introduced to provide a geometric link between end-effector configurations and the tip of the rod (see Figure 1).

### Proposed Approach

This simulation-based research practice builds upon the Cosserat deformable rod model and addresses the problem of improving simplified models for deformable linear objects (DLOs) through feedback-based correction. The main goal is to reduce the mismatch between simplified and high-fidelity representations in robotic manipulation settings.

The proposed approach is inspired by the closed-loop shape control framework introduced in [2], which highlights the importance of feedback mechanisms in deformable object manipulation. In contrast to perception-based feedback, this work relies on a model-based mismatch to drive the correction process. A simplified model (e.g., mass-spring) is used to generate an initial actuation through a feedforward optimization step, and the resulting deformation is evaluated using a high-fidelity reference model (e.g., Cosserat rod model).

A feedback mechanism is introduced to iteratively correct the actuation by propagating shape errors from the reference model back to the simplified model. This correction relies on a local mapping between actuation and shape variations, enabling progressive refinement of the predicted deformation while preserving computational efficiency.

As a possible extension, perception-based feedback and robotic coupling can be considered to enable physical implementation.

### References

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